SARANG RAJ

Kozhikode, Kerala, India | +91 73066 69609 | sarangraj188@gmail.com
github.com/SarangRaj01 | leetcode.com/u/Sarang_Raj

SUMMARY

Computer Science undergraduate at VIT Vellore (CGPA 9.40) with hands-on experience in backend development with Java and Spring Boot, relational database design in MySQL, systems programming in C, and socket-level networking in Python. Solved 76 LeetCode problems, built a thread-safe inventory system with automated concurrency tests, and presented a cloud computing talk to an audience of 60+ students.

EDUCATION

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu — Expected 2028
B.Tech in Computer Science and Engineering | CGPA: 9.40 (3rd Year)

Kendriya Vidyalaya No. 1 — 2021 – 2023
Class XII (CBSE): 81.8% | Class X (CBSE): 91.2%

TECHNICAL SKILLS

- **Languages:** Java, Python, C, C++, JavaScript, SQL
- **Backend and Web:** Spring Boot, Spring Data JPA, Hibernate, REST APIs, React, Vite, HTML, CSS
- **Databases:** MySQL, relational schema design, transactions, row-level locking
- **Tools and Platforms:** Git, GitHub, Docker, Maven, Postman, Wireshark, LLVM/Clang, GCC, MS Excel
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS, Compiler Design, Cloud Computing

PROJECTS

Distributed Inventory and Order Consistency System | *Java 21, Spring Boot, Hibernate/JPA, MySQL, React, Maven* | github.com/SarangRaj01/ProjectDBMS

- Built a REST backend with 11 endpoints over a 12-table MySQL schema covering products, warehouses, inventory, customers, orders, and stock reservations across multiple warehouses.
- Prevented overselling under concurrent orders using pessimistic row-level locking (SELECT ... FOR UPDATE), JPA @Version optimistic checks, and a database CHECK constraint (reserved quantity <= on-hand quantity).
- Implemented atomic multi-item order placement and cancellation with @Transactional rollback, idempotency keys to block duplicate submissions, and audit tables for inventory movements and order status history.
- Wrote 16 automated unit, WebMvc, and multi-threaded tests using CyclicBarrier; two simultaneous requests for 7 units from 10 in stock result in exactly one success and no oversell.
- Built a React and Vite dashboard showing live stock levels and an order placement flow.

Self-Healing Compiler (SHC): Runtime Vulnerability Detection and Mitigation | *C11, Python, GCC/Clang* | github.com/SarangRaj01/self-healing-compiler

- Capstone project for BCSE307P Compiler Design Lab (team of 3): a compiler toolchain that injects shadow-stack, bounds-check, and control-flow-integrity probes and pre-builds alternate function variants, so a running service can recover from memory-corruption exploits instead of crashing.
- Reported 100% post-attack availability and about 4.2 microseconds detect-to-heal latency, versus 300 to 1800 seconds for a rebuild and redeploy, with under 6.5% overhead on a benign workload and 0 false positives across 1,500,000 probes.
- My contributions: problem identification, a 15-paper literature survey, and design of the exploit benchmarks (a 180-byte buffer-overflow packet against a 64-byte stack frame, and an indirect function-pointer hijack).

LAN Messenger with Network Discovery | *Python, TCP/UDP Sockets, Wireshark*

- Built a LAN chat and multiplayer-style lobby from scratch: a TCP client-server chat extended with UDP broadcast discovery so peers find each other without typing IP addresses.
- Designed a discovery protocol where each machine broadcasts a "who's here" message and listens for replies, populating a live list of peers with their IP and port.
- Tested across multiple laptops on one Wi-Fi network and traced traffic through the Ethernet, IP, TCP, and UDP layers with Wireshark captures and ARP-based MAC resolution.

Real-Time Chat Application | *Python, TCP Sockets, Multithreading*

- Built a multi-client chat server on raw TCP sockets that runs one thread per connection, so many users can chat at once.
- Handled client join and leave events and broadcast messages to all connected clients in real time.

ACHIEVEMENTS AND LEADERSHIP

- **Technical talk on cloud computing:** Presented a topic on cloud computing to an audience of 60+ students.
- **LeetCode (Sarang_Raj):** 76 problems solved ;123 submissions in the past year across 34 active days; contest rating 1,490.